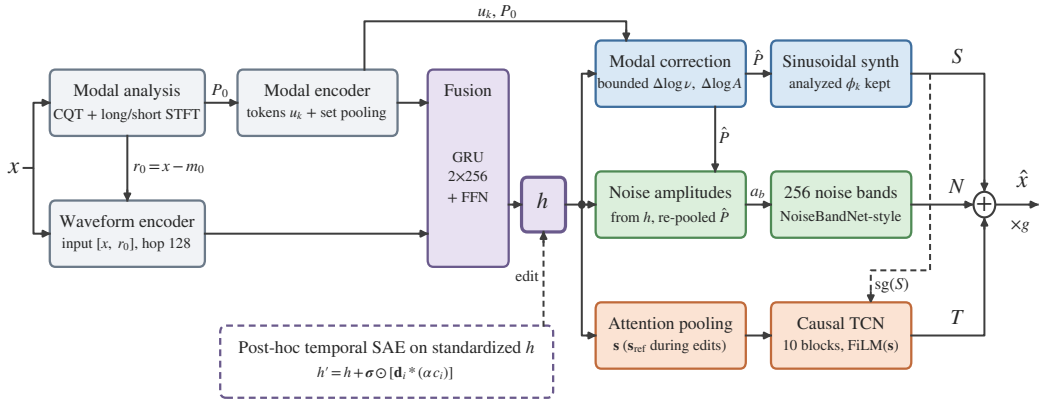
u_k, P_0
Modal analysis
CQT + long/short STFT
P_0
Modal encoder
tokens u_k + set pooling
Fusion
GRU
2×256
+ FFN
h
edit
x
$r_0 = x - m_0$
Waveform encoder
input [x, r_0], hop 128
Post-hoc temporal SAE on standardized h
$h' = h + \sigma \odot [\mathbf{d}_i * (\alpha c_i)]$
Modal correction
bounded $\Delta \log \nu$, $\Delta \log A$
$\hat{P}$
Sinusoidal synth
analyzed ϕ_k kept
S
$\hat{P}$
Noise amplitudes
from h, re-pooled $\hat{P}$
a_b
256 noise bands
NoiseBandNet-style
N
$\hat{x}$
$\times g$
sg(S)
Attention pooling
s ($\mathbf{s}_{\text{ref}}$ during edits)
Causal TCN
10 blocks, FiLM(s)
T